7. 6 Regulation of cardiac activity

The type of heart in human is myogenic because the heart beat originates from the muscles of the heart. The nervous and endocrine systems work together with paracrine signals (metabolic activity) to influence the diameter of the arterioles and alter the blood flow. The neuronal control is achieved through autonomic nervous system (sympathetic and parasympathetic). Sympathetic neurons release **nor-epinephrine** and adrenal medulla releases **epinephrine**. The two hormones bind to β – adrenergic receptors and increase the heart rate. The parasympathetic neurons secrete acetylcholine that binds to **muscarinic receptors** and decreases the heart beat. **Vasopressin** and **angiotensin II**, involved in the regulation of the kidneys, results in vasoconstriction while **natriuretic peptide** promotes vasodilation. Vagus nerve is a parasympathetic nerve that supplies the atrium especially the SA and the AV nodes.

7.7 Disorders of the circulatory system

Hypertension

Hypertension is the most common circulatory disease. The normal blood pressure in man is 120/80 mmHg. In cases when the diastolic pressure exceeds 90 mm Hg and the systolic pressure exceeds 150 mm Hg persistently, the condition is called hypertension. Uncontrolled hypertension may damage the heart, brain and kidneys.

Coronary heart disease

Coronary heart disease occurs when the arteries are lined by **atheroma**. The build-up of atheroma contains cholesterol, fibres, dead muscle and platelets and is termed **Atherosclerosis**. The cholesterol rich atheroma forms plaques in the inner lining of the arteries making them less elastic and reduces the blood flow. Plaque grows within the artery and tends to form blood clots, forming coronary **thrombus**. Thrombus in a coronary artery results in **heart attack**.

Stroke

Stroke is a condition when the blood vessels in the brain bursts, **(Brain haemorrhage)** or when there is a block in the artery that supplies the brain, (atherosclerosis) or thrombus. The part of the brain tissue that is supplied by this damaged artery dies due to lack of oxygen **(cerebral infarction)**.

Angina pectoris

Angina pectoris (ischemic pain in the heart muscles) is experienced during early stages of coronary heart disease. Atheroma may partially block the coronary artery and reduce the blood supply to the heart. As a result, there is tightness or choking with difficulty in breathing. This leads to **angina** or chest pain. Usually it lasts for a short duration of time.

Myocardial infarction (Heart failure)

The prime defect in heart failure is a decrease in cardiac muscle contractility. The Frank- Starling curve shifts downwards and towards the right such that for a given EDV, a failing heart pumps out a smaller stroke volume than a normal healthy heart.

When the blood supply to the heart muscle or myocardium is remarkably reduced it leads to death of the muscle fibres. This condition is called **heart attack**

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